

AN INTEGRATED SOUND VELOCITY PROFILING PROBE

ARCHITECTURE FOR CLOSED-LOOP ACOUSTIC CALIBRATION, ADAPTIVE SAMPLING, AND SPATIAL CONFIDENCE ESTIMATION

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ABSTRACT DISCLOSURES

Sound velocity profiling is fundamental to accurate acoustic depth measurement, yet conventional practice treats profile acquisition, system calibration, and quality control as disconnected operations. This paper discloses a methodology for continuous closed-loop validation of sonar depth against mechanical measurement through an integrated probe architecture. The system implements thermal-lag compensation through one or more timing, modeling, or sensor-fusion strategies, and adaptive sampling triggered by real-time sensor discrepancy detection. A spatial confidence framework is disclosed that optimizes the trade-off between survey efficiency and data quality through dynamic interpolation weighting and boundary-calibrated uncertainty estimation. The methodology is hardware-agnostic, applicable across freshwater and marine environments with high spatial or temporal velocity gradients, and intended for integration with any hydrographic, bathymetric, or acoustic survey platform, whether crewed, autonomous, towed, or fixed. The methodology is disclosed in broad, hardware-agnostic terms herein, with a concrete illustrative implementation provided in a companion document.

1. INTRODUCTION

Sound velocity is the primary environmental variable governing the accuracy of acoustic depth measurements in hydrographic surveying. Temperature, salinity, and pressure gradients along the water column create velocity variations that refract acoustic rays; without correction, these variations introduce systematic depth errors that propagate through the entire bathymetric product.

Standard practice for sound velocity profiling employs dedicated instruments — typically time-of-flight, time-of-travel, or acoustic-resonance probes that measure sound velocity alongside temperature and pressure, directly or by derivation from other measured properties. Concurrently, bar-check or equivalent mechanical-reference procedures provide independent validation of sonar depth calibration, using a physical target lowered, extended, or otherwise positioned at known depths beneath or near the transducer.

These operations are conventionally performed as separate manual tasks. Neither informs the other in real time, and neither adapts its sampling strategy based on the other's measurements.

This paper discloses a methodology that closes this loop: an integrated probe or probe system that measures sound velocity while simultaneously providing, or coordinating with, a physical or synthetic reference target for the sonar. The system enables real-time cross-validation and adaptive sampling in response to detected gradients. A spatial confidence framework is disclosed that quantifies uncertainty across a survey area.

2. SCOPE

This paper presents a conceptual architecture and set of operational methods for integrated sound velocity acquisition, mechanical or synthetic depth reference, adaptive sampling, and spatial/temporal uncertainty management, in any combination or subset. Specific implementations of mechanical design, electronics, firmware, communications, sensor selection, correction algorithms, and deployment platform are intentionally disclosed as open design choices, as the overarching architecture encompasses any equivalent instantiation. Any system that implements one or more of the closed-loop, adaptive, or confidence-mapping principles described herein — regardless of the specific sensors, formulas, thresholds, timing schemes, or hardware used to do so — falls within the scope of this disclosure. A practical, concrete implementation of one configuration is detailed in the companion document published alongside this framework.

3. PROBLEM STATEMENT

3.1 Thermal and Response-Lag Artifacts

Commercial sound velocity probes achieve high static accuracy under equilibrium conditions. However, these specifications assume the sensor has reached thermal or otherwise physical equilibrium with its environment. Any sensor package with nonzero thermal mass, response time, or signal-propagation delay will, under a moving deployment through a gradient (thermal, salinity, pressure, or otherwise), log a value that lags the true instantaneous condition at its physical position. This lag is rarely addressed systematically in hydrographic operational protocols, as distinct from post-processing corrections used elsewhere in oceanographic practice. The resulting error is systematic and depth- or gradient-dependent, and may be addressed through any combination of controlled dwell/soak timing, active or passive thermal conditioning, real-time or post-hoc lag modeling, sensor fusion with a faster-responding proxy sensor, or equivalent methods.

3.2 Fixed-Increment Sampling and Missed Gradients

Conventional SVP profiles are collected at fixed depth or time increments regardless of actual gradient structure. Any sharp gradient layer lying between two sampling points is interpolated across, introducing error proportional to gradient strength and inter-sample distance. This may be addressed through adaptive, event-triggered, or feedback-driven resampling of any kind, including but not

limited to increment reduction, deployment-rate modulation, or supplemental targeted casts.

3.3 The Manual Bar-Check Bottleneck

Bar-check and equivalent mechanical-reference procedures remain manual, are typically performed once per survey session, and are treated as a static calibration, despite sound velocity conditions evolving continuously through diurnal heating, mixing, or stratification. This may be addressed through continuous, periodic, or event-triggered mechanical-acoustic cross-validation performed throughout a survey rather than as a single discrete event, using any physical, synthetic, or virtual reference target.

3.4 Spatial and Temporal Uncertainty

A single SVP cast represents ground truth only at its coordinate and timestamp; confidence in that measurement decays with distance and elapsed time by some function — deterministic, stochastic, geostatistical, or learned — that is rarely made explicit or updated during a survey. A methodology that integrates a continuously updated confidence field directly into adaptive acquisition and survey planning, in real time, during the survey itself, remains absent from standard practice and is disclosed herein in general form.

4. PROPOSED ARCHITECTURE: INTEGRATED SOUND VELOCITY PROBE

4.1 Combined-Function Design

The disclosed architecture integrates, within a single housing, multiple co-located housings, or a coordinated multi-body system, two or more of the following functions:

- An SVP payload, comprising a sound-velocity-related sensor (time-of-flight, time-of-travel, resonance-based, or derived from temperature/pressure/salinity) with co-located or fused temperature and pressure sensing.
- A mechanical, acoustic, optical, or synthetic reference target providing a detectable return or reference point for independent depth validation.

This integration eliminates the operational overhead of separate deployments for velocity profiling and depth-reference calibration, regardless of the specific sensing modality or target type used, and regardless of whether the target and sensor package share a housing or are mechanically coupled by any other fixed or computable geometric relationship.

4.2 Redundant Depth Measurement

The probe or probe system incorporates one or more redundant, non-acoustic and/or non-pressure-dependent means of establishing physical position or displacement, including but not limited to: rotary or linear encoders, physical fiducials or markers, winch or spool telemetry, inertial or dead-reckoning estimation, optical tracking, or any combination thereof, with or without correction for cable angle, catenary, current-induced deflection, or platform motion. This redundant

reference is used to continuously validate acoustic depth measurement against a physically independent measurement.

4.3 Lag Compensation

The disclosed methodology may employ any one or combination of the following, non-exhaustively: (a) a controlled-rate deployment with programmed dwell or pause intervals sufficient for sensor equilibration, empirically or analytically determined; (b) real-time or post-hoc mathematical lag correction based on a modeled or measured sensor response function; (c) sensor fusion with a lower-lag proxy signal; (d) active thermal or physical conditioning of the sensor package. The specific dwell duration, model parameters, or fusion weighting are implementation-specific and are not disclosed herein, as they are readily determined by a practitioner through empirical characterization of the chosen sensor package.

4.4 Adaptive Sampling

The system implements a closed-loop sampling strategy, in which one or more of the following occur, in any order or combination, within a single cast or across multiple casts:

- 1 Profiling proceeds at a default increment, rate, or schedule.
- 2 At one or more depths or times, an acoustic-derived depth or velocity-related measurement is compared against an independent mechanical or reference measurement.
- 3 Where the discrepancy exceeds a configurable, adaptive, or learned tolerance, the corresponding interval is flagged as requiring higher-resolution characterization.
- 4 Sampling parameters (increment, rate, dwell time, or deployment velocity) are adjusted — either immediately within the same cast, or in a subsequent cast at the same or a nearby coordinate — to resolve the flagged interval at higher resolution.

This targets sampling effort toward regions of high gradient or high discrepancy by any triggering logic and at any response timescale.

5. DESIGN PHILOSOPHY

- **Hardware Independence.** The methodology is defined by operational logic, not by any specific sensor, actuator, platform, or algorithm.
- **Continuous Validation.** Acoustic measurements are cross-validated against an independent reference continuously or at high frequency, rather than through periodic manual calibration events, using any comparison, fusion, or statistical method.
- **Explicit Uncertainty.** Confidence or uncertainty is treated as an explicit, mapped, and dynamically updated quantity by any modeling approach, rather than a hidden or purely post-processed artifact.

- **Active Learning.** A subset of sampling effort is allocated to validating the uncertainty model itself against reality, with the model and/or sampling plan updated in response, by any adaptive or learning method.
- **Traceable Audit Trail.** Measurements are logged with synchronized time and associated reference, environmental, and confidence metadata, by any data architecture, for retrospective quality assessment.

6. SPATIAL AND TEMPORAL CONFIDENCE FRAMEWORK

6.1 *Uncertainty as a Mapped Quantity*

The disclosed framework defines a confidence field over a survey area and/or time window, with confidence at a measurement's own coordinate/time set to a maximum value and decaying, by any deterministic, stochastic, geostatistical, or learned decay function (including but not limited to exponential, Gaussian, power-law, kriging-based, or machine-learned models), with distance, time, or any other relevant covariate. Confidence at any unmeasured point may be computed by interpolation, extrapolation, or model inference from one or more surrounding measurements, by any method.

6.2 *Boundary or Covariance Sampling*

A subset of sampling effort may be allocated to regions between or at the edges of primary sampling cells, specifically to empirically measure spatial or temporal covariance/gradient and calibrate the parameters of the confidence decay model in situ, by any measurement or estimation method, with the survey or sampling plan adjusted in response in real time or after the fact.

6.3 *Temporal Decoupling*

Survey geometry may incorporate multiple passes, headings, or revisit times to capture temporal evolution of the velocity field, with measurements from different times associated with, weighted toward, or excluded from specific portions of the final product by any temporal-relevance criterion.

7. OPERATIONAL PROTOCOL (GENERALIZED)

7.1 *Cast Sequence*

A cast may include, in any order or combination: positioning at a coordinate; controlled deployment; one or more paused or continuous sampling events comprising velocity-related, mechanical-reference, and acoustic measurements; comparison of acoustic and mechanical/reference depth with flagging of discrepancies exceeding a tolerance; and retrieval, during which additional data may optionally be collected and cross-referenced against the deployment-phase data to detect temporal changes in the water column.

7.2 *Closed-Loop Response*

Where a discrepancy or gradient is flagged, the system may respond by any combination of: an immediate in-cast adjustment of sampling parameters; scheduling of a supplemental cast at the same or nearby coordinate; or downstream replacement of interpolated values with higher-resolution measured data for the flagged interval. The response trigger and response type are configurable and are not limited to any specific threshold, timing, or method.

8. SYSTEM INTEGRATION

8.1 Platform Compatibility

The methodology may be integrated with any survey platform — crewed vessel, autonomous surface or underwater vehicle, towed body, fixed installation, or handheld system — and is not limited to any specific water body type, depth range, or deployment method. Its value lies in the operational and data-fusion logic rather than in any specific hardware implementation.

8.2 Data Architecture

The system maintains a timestamped data record, structured by any architecture (including but not limited to append-only ledgers, relational, or streaming formats), associating each measurement with synchronized time, mechanical/reference depth, acoustic depth, velocity-related environmental data, and confidence metadata, communicated by any protocol and processed at any point in the pipeline (edge, onboard, or post-mission).

9. DISCUSSION

9.1 Operational Efficiency

By combining functions conventionally performed as separate manual operations, and by targeting supplemental sampling only where discrepancy or gradient is detected, the disclosed methodology may reduce total station time, deployment count, or energy expenditure relative to conventional practice, with the magnitude of benefit dependent on gradient frequency and site conditions.

9.2 Data Quality and Auditability

Continuous mechanical-acoustic cross-validation, combined with an explicit confidence framework, shifts quality assurance from purely post-processed statistical estimation toward real-time, traceable verification, supporting a range of survey order or engineering-grade data requirements.

10. LIMITATIONS AND KNOWN FAILURE MODES (ADDRESSED GENERALLY)

A complete implementation of this architecture must contend with a number of known failure modes, each of which admits multiple independent classes of solution and is not limited to any single approach disclosed here:

- Reference-target detectability under variable turbidity, biofouling, multipath, or acoustic clutter — addressable through target material/geometry selection, signal processing, or redundant target modalities.
- Reference-path geometry error arising from cable angle, catenary, current deflection, or platform motion — addressable through angular sensing, geometric correction modeling, active tensioning, or platform-relative compensation.
- Sensor package response characteristics varying by housing material, size, and deployment rate — addressable through empirical characterization, active conditioning, or fusion-based correction, per Section 4.3.
- Confidence model misspecification in heterogeneous or non-stationary water bodies — addressable through model class selection, online recalibration, or ensemble methods, per Section 6.

11. CONCLUSION

This paper discloses a conceptual architecture and methodology for integrated, closed-loop sound velocity profiling, mechanical-acoustic depth validation, adaptive sampling, and explicit spatial/temporal confidence estimation, applicable across a broad range of hardware, platforms, and environments. Specific implementations, algorithms, and parameters are intentionally left unspecified and are understood to be readily determinable by a practitioner skilled in hydrographic survey systems, marine sensor design, or related fields, without undue experimentation, given the operational logic disclosed herein.

12. REFERENCES

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INTELLECTUAL PROPERTY NOTICE / PRIOR ART DISCLOSURE

This document constitutes a public, defensive disclosure of a conceptual methodology and system architecture, made available to establish prior art. The functional logic, adaptive sampling triggers, closed-loop calibration methods, lag-compensation strategies, and uncertainty frameworks described herein — including all named and unnamed equivalent methods falling within the general classes disclosed in each section — are intended to remain in the public domain as prior art, to prevent the issuance of patents restricting practice of these methods by any party, regardless of the specific implementation, sensor, algorithm, or parameter values used.

Publication date: July 2026. This disclosure is intended for timestamped public archival (e.g., arXiv, TechRxiv, or a defensive publication registry) to establish an independently verifiable priority date.

A companion document providing a concrete, illustrative technical description of one implementation within this architecture family is published separately under the same title (see companion repository document). That document serves as the enabling disclosure, demonstrating the practical realization of the framework disclosed herein.